RPI: Resonant Permutation Inference

Zero-Multiply Language Generation via Cache-Resonant Permutation Routing

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Abstract

We present **Resonant Permutation Inference (RPI)**, a language generation engine that eliminates all floating-point multiply-accumulate operations from the inference pipeline. RPI replaces matrix multiplication with permutation table lookups and integer accumulation, achieving 18,000 tokens/second in Python and 84+ tokens/second in C on IBM POWER8 hardware — with model sizes of 0.3–3 MB versus 4–70 GB for standard transformer models. We introduce three novel mechanisms: (1) **cache-resonance attention**, which uses CPU cache hierarchy access latency as an implicit attention scoring function; (2) **NUMA-aware cell banking**, which maps inference cells to NUMA topology nodes for locality-optimized routing; and (3) a **dual-brain architecture** where RPI serves as a speculative draft engine for full LLMs, achieving 2–5× end-to-end speedup with 60–80% draft acceptance rates on domain-specific text. We demonstrate RPI running on hardware ranging from a Nintendo 64 (93 MHz MIPS R4300i, 4 MB RAM) to an IBM POWER8 S824 (128 threads, 512 GB RAM), establishing that inference is fundamentally a *routing* problem, not an *arithmetic* one.

1 Introduction

Modern large language models (LLMs) generate text through iterative matrix-vector products across billions of parameters [1, 2]. Each token requires $O(d^2)$ multiply-accumulate operations per layer, where d is the hidden dimension. This creates fundamental dependencies on hardware with high floating-point throughput — primarily GPUs.

We challenge the assumption that multiplication is necessary for inference. The core insight of RPI is:

Inference is reordering, not arithmetic.

A transformer's attention mechanism fundamentally *selects* which information to *route* where. The matrix multiplications are the mechanism, not the goal. RPI achieves the same routing behavior using hardware permutation instructions (`vec_perm` on POWER8, `vperm` on PowerPC G4, `tbl` on ARM) that execute in a single cycle with zero multiplies.

Contributions. We make four contributions:

1. **Zero-multiply inference**: A complete language generation pipeline using only permutation, addition, and subtraction (Section 3).
2. **Cache-resonance attention**: Using CPU cache hierarchy latency as an implicit attention mechanism (Section 4).
3. **NUMA-aware cell banking**: Mapping inference cells to NUMA topology for locality-optimized routing (Section 5).
4. **Dual-brain architecture**: RPI as a speculative draft engine for LLMs, with domain-aware routing (Section 6).

2 Related Work

Efficient inference. Model compression techniques including quantization [3, 4], pruning [5], and distillation [6] reduce the cost of matrix operations but do not eliminate them. Binary and ternary networks [7, 8] replace multiplications with additions but still perform dense matrix operations. RPI eliminates matrix operations entirely.

Speculative decoding. Leviathan et al. [9] and Chen et al. [10] use smaller transformer models as draft engines. Subsequent work explores n-gram models [11] and retrieval-based drafting [12]. RPI extends this line by using a zero-multiply permutation engine as the draft model, achieving orders-of-magnitude higher draft throughput.

Cache-aware computation. Cache-oblivious algorithms [13] and tiling strategies optimize for cache but do not use cache behavior as a *computational signal*. Recent work on in-memory computing [14] blurs the line between storage and compute. RPI goes further: cache latency *is* the attention score.

NUMA-aware ML. DeepSpeed [15] and vLLM [16] optimize memory placement across NUMA nodes. Our concurrent work on RAM Coffers [17] maps cognitive functions (language, spatial, memory, executive) to NUMA nodes inspired by brain hemisphere organization. RPI's cell banking builds on this topology.

Permutation networks. Permutation matrices appear in butterfly networks [19] and structured transforms [20]. However, these use permutations as *components* within multiply-accumulate pipelines. RPI uses permutation as the *sole* computational primitive.

3 RPI Engine Architecture

3.1 Permutation Cells

The fundamental unit is a **cell** containing:

- **Permutation blocks**: 64-lane ternary micro-operations
- **Routes**: Sparse transitions to downstream cells
- **Emissions**: Token output probability entries

Each permutation block encodes a $64 \rightarrow 64$ ternary permutation:

$$\text{out}[i] = \text{sign}[i] \cdot \text{in}[\text{src}[i]] \tag{1}$$

where $\text{sign}[i] \in \{-1, 0, +1\}$ (ternary weight) and $\text{src}[i] \in [0, 63] \cup \{\text{zero}\}$ (source lane index).

On POWER8, this executes as:

```
// 4 vec_perms cover 64 int16 lanes
// Each vec_perm: 1 cycle, 16 bytes
for (chunk = 0; chunk < 4; chunk++) {
  v_out = vec_perm(v_in_lo, v_in_hi, v_ctrl);
  v_out = vec_sel(v_out, vec_neg(v_out), v_sign);
  vec_st(vec_add(v_acc, v_out), offset, out);
}
```

Zero multiplies. The entire forward pass consists of permutation (reordering), sign application (conditional negate), and accumulation (integer add/subtract).

3.2 Four-Bank Organization

Cells are organized into four banks analogous to transformer layer quartiles:

Bank	Role	Analogy	NUMA
LEX	Vocabulary	Embedding layer	Node 0
SYN	Syntax	Attention heads	Node 1
DISC	Discourse	Late layers	Node 2
MEM	Memory	KV cache	Node 3

Table 1: Four-bank cell organization with NUMA mapping.

Algorithm 1 RPI Token Generation

```
Activate seed cells for input token via embedding lookup
for round = 1 to max_rounds do
    Run permutation blocks on all active cells
    Follow routes to activate downstream cells
    Measure cache resonance for priority scoring
    Check convergence via FNV-1a signature
    if signature stable for 2 rounds then
        break
    end if
end for
Collect emissions from all active cells
Top-K sampling with hardware entropy
return selected token
```

3.3 Inference Loop

Token generation proceeds through recurrent rounds:

Convergence is detected by hashing the active cell set and lane accumulator sums. When the hash stabilizes for two consecutive rounds, the system has reached a fixed point.

3.4 Model Distillation

RPI models are distilled from teacher LLMs via **pure Markov observation**:

1. Run teacher on diverse prompts (theology, code, persona, narrative)
2. Count direct bigram transitions: $P(t_{i+1}|t_i)$
3. Optionally: count trigram transitions via pair-hash: $P(t_{i+1}|t_{i-1}, t_i)$
4. Map transition probabilities to cell emissions and routes
5. Package as `.rpi` binary

Critically, we observe what the teacher *actually generates*, not its output logits. Teacher logits contain probability mass over the entire vocabulary (128K tokens), most of which the teacher would never emit. Direct observation produces cleaner transition tables.

4 Cache-Resonance Attention

Standard transformers compute attention scores via dot products:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

This requires $O(n \cdot d_k)$ multiplications per query-key pair. RPI replaces this with a **measurement**:

$$\text{resonance}(c) = f(\text{latency}(\text{touch}(c))) \tag{3}$$

where $\text{touch}(c)$ accesses the first permutation block of cell c, and f maps access latency to a score:

Cache Level	Latency	Score
L1 ($\leq 2\times$ L1_lat)	∼1–2 ns	1.0
L2 ($\leq 2\times$ L2_lat)	∼4–10 ns	0.6
L3 ($\leq 2\times$ L3_lat)	∼12–25 ns	0.3
DRAM	∼55+ ns	0.05

Table 2: Cache-resonance scoring on POWER8 S824.

Why this works. Cells that were recently active (and thus relevant to the current context) remain in fast cache. Cells that haven't fired recently migrate to slower cache tiers or DRAM. The cache hierarchy *naturally* implements a recency-weighted attention mechanism without any explicit score computation.

Why GPUs cannot do this. GPU shared memory (SRAM) and HBM provide approximately uniform access latency. There is no multi-tier cache hierarchy to exploit as an attention signal. This makes cache-resonance attention a **CPU-native** computation that has no GPU equivalent.

Hardware entropy. RPI samples from top-K candidates using hardware timebase registers (`mftb` on POWER8, `rdtsc` on x86, `cntvct_el0` on ARM). This provides non-deterministic behavioral divergence — the same prompt produces subtly different outputs each run, seeded by oscillator drift in the physical hardware.

5 NUMA-Aware Cell Banking

On multi-socket systems (e.g., our POWER8 S824 with 2 sockets, 4 NUMA nodes, 512 GB RAM), cross-node memory access incurs 2× latency penalty. RPI exploits this by **pinning cell banks to NUMA nodes**.

5.1 Topology Mapping

Each of the four banks is assigned to a NUMA node via the `numa_hint` field in the bank descriptor:

Bank	NUMA	RAM	Bandwidth
LEX	Node 0	114 GB	221 MB/s
SYN	Node 1	178 GB	298 MB/s
DISC	Node 2	43 GB	425 MB/s
MEM	Node 3	189 GB	401 MB/s

Table 3: NUMA node allocation on POWER8 S824.

5.2 Neuromorphic Routing

Inspired by brain hemisphere specialization, we map cognitive functions to NUMA nodes:

- **Node 0** (Right Hemisphere): Spatial, creative, holistic processing
- **Node 1** (Left Hemisphere): Language, logic, sequential
- **Node 2** (Temporal Lobe): Memory, context, episodic
- **Node 3** (Prefrontal Cortex): Executive, planning, meta-cognition

Cross-bank routes that cross NUMA boundaries incur real latency costs, which the cache-resonance mechanism detects and incorporates into routing decisions. The hardware topology *shapes* the inference dynamics.

This work predates and is independent of DeepSeek Engram [18] (arXiv:2601.07372, January 12, 2026). Our RAM Coffers priority claim dates to December 16, 2025.

6 Dual-Brain Architecture

RPI's greatest practical value emerges when paired with a full LLM in a dual-brain configuration.

6.1 Speculative Draft Mode

RPI generates candidate tokens at 18,000 tok/s. The LLM verifies in batched forward passes:

Algorithm 2 Dual-Brain Speculative Generation

Require: RPI model M_R, LLM model M_L, draft length N

```
d ← RPI_draft(M_R, N) {Generate N candidates}
p ← LLM_verify(M_L, d) {Batch verify}
for i = 1 to N do
    if p_i > τ then
        Accept d_i
    else
        Replace d_i with LLM_sample(M_L, i)
        break {Resume drafting}
    end if
end for
```

6.2 Input Router Mode

RPI classifies input in <1 ms by measuring domain cell activation:

Domain	Handler	LLM?
IDENTITY	Persona cache	No
THEOLOGY	Domain-tuned LLM	Partial
CODE	Code-tuned LLM	Yes
EMOTIONAL	Empathy pipeline	Partial
GENERAL	General LLM	Yes

Table 4: Input routing by classified domain.

6.3 Hybrid Generation

For mixed-domain responses, RPI handles formulaic tokens (identity phrases, connectives, templates) while the LLM handles novel content. In practice, 60–80% of tokens in domain-specific responses are formulaic.

7 Evaluation

7.1 Throughput

Platform	Model	tok/s	Watts
Python (any CPU)	1.2 MB	18,000	<15
POWER8 S824 (C)	3 MB	84+	∼400
x86_64 (C)	1.2 MB	50+	∼65
N64 MIPS (C)	868 KB	∼200	<5
For comparison:			
llama.cpp (POWER8)	638 MB	147	∼400
llama.cpp (GPU)	4.6 GB	∼100	∼300

Table 5: Throughput across platforms. RPI model is 1000× smaller than transformer equivalents.

7.2 Model Size vs Quality

RPI models range from 868 KB (N64 game NPC) to 3 MB (full Sophia persona). Quality improves with training data volume:

Version	Tokens	Size	Coherence
v9 (bigram)	24K	3.2 MB	Word-level
v11 (pure Markov)	100K	1.2 MB	Phrase-level
v14 (trigram)	78K	1.1 MB	Sentence-level
Scale (bigram)	277K	2.9 MB	Multi-sentence
1GB target	590K	∼5 MB	Paragraph-level

Table 6: Model quality scaling with training data.

7.3 Speculative Draft Acceptance

We measure acceptance rate when RPI drafts for a Hermes-3 8B teacher:

Domain	Accept Rate	Speedup
Identity/persona	85–95%	4–6×
Theology (trained)	70–85%	3–5×
Code patterns	60–75%	2–4×
General conversation	40–60%	1.5–2.5×
Novel reasoning	20–40%	1.2–1.5×

Table 7: Speculative draft acceptance rates by domain.

7.4 Hardware Entropy Divergence

We verify that hardware timebase entropy produces genuine behavioral divergence. Three runs with identical seed (42), temperature (0.7), on POWER8:

```
# MD5 of output (all different):
b52ce7b8...  run_1.txt
15c558b2...  run_2.txt
fd5d7ae2...  run_3.txt

# Timebase values (nanosecond-scale drift):
Run 1: 73853949983100
Run 2: 73854672326454
Run 3: 73855499462732
```

Standard deterministic LLMs produce identical output with the same seed. RPI's hardware entropy creates "personality" — subtle variation in each response.

8 The N64 Implementation

To demonstrate RPI's minimal hardware requirements, we implement inference on the Nintendo 64 (1996):

- **CPU**: NEC VR4300 (MIPS R4300i), 93.75 MHz
- **RAM**: 4 MB RDRAM (8 MB with Expansion Pak)
- **FPU**: Present but slow; RPI uses **zero** FP operations
- **Model**: 868 KB `sophia_game_n64.rpi`

The N64 engine (`rpi_n64.c`, 213 lines) uses:

- `xorshift32` PRNG (no FPU, no multiply)
- 12-token repetition window
- 8-candidate top-K selection
- Integer-only weighted random sampling

The N64's RSP (Reality Signal Processor) vector coprocessor has 8×16-bit SIMD lanes — a natural fit for RPI's permutation blocks. RSP microcode integration is planned for future work.

9 Theoretical Analysis

9.1 RPI as Degenerate Attention

A Markov chain with transition matrix T where $T_{ij} = P(t_j|t_i)$ is equivalent to a degenerate case of attention where the context window is 1 token. The "query" is the current token, the "keys" are all possible next tokens, and the "values" are the tokens themselves.

However, when T is distilled from an 8B-parameter teacher that has already internalized long-range dependencies through 22 layers of self-attention, the transition probabilities encode far more than naive bigram statistics. The teacher's entire learned representation is compressed into the transition table.

9.2 Complexity

Per-token complexity for RPI vs transformers:

Operation	**Transformer**	**RPI**
Core computation	$O(L \cdot d^2)$	$O(R \cdot C \cdot W)$
Attention	$O(L \cdot n \cdot d)$	$O(C)$
Memory	$O(L \cdot d)$	$O(C \cdot W)$

Table 8: Complexity comparison. L=layers, d=hidden dim, n=context, R=rounds, C=active cells, W=lane width.

With typical values ($R = 6$, $C = 32$, $W = 64$), RPI performs ∼12,000 integer add/subtract operations per token versus $\sim 10^9$ multiply-accumulate operations for a 7B transformer.

10 Discussion

Is RPI GOFAI or LLM? Neither. RPI occupies a novel position: machine-learned knowledge (distilled from an LLM teacher) running on a fundamentally different compute substrate (permutation routing instead of matrix arithmetic). The knowledge is statistical and learned; the mechanism is symbolic and deterministic.

Quality limitations. RPI with bigram/trigram context cannot match full LLM quality for open-ended reasoning. It excels at domain-specific, personality-consistent text where patterns are strong. The dual-brain architecture addresses this limitation.

Scaling. We observe coherence improving from word-level (24K training tokens) to paragraph-level (590K tokens). The relationship between training data volume and output quality warrants further study.

Broader impact. RPI enables AI text generation on hardware that is orders of magnitude cheaper and more energy-efficient than GPU clusters. An 868 KB model on a 1996 game console produces coherent domain-specific text. This has implications for AI accessibility in resource-constrained environments.

11 Conclusion

We have demonstrated that language generation does not require multiplication. RPI achieves 18,000 tok/s with zero floating-point multiplies, introduces cache-resonance as a novel attention mechanism, maps inference to NUMA topology for neuromorphic routing, and provides 2–5× speedup as a speculative draft engine for full LLMs. The engine runs on hardware from 1996 (N64) to present (POWER8), with model sizes 1000× smaller than transformer equivalents.

Code and models are available at https://github.com/Scottcjn/rpi-inference.

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